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What is Linux?

Linux is an Operating System (OS)

OS means: software that controls computer hardware and runs programs

Just like:

- Windows
- macOS

But Linux is:

- Free (open-source)
- Highly secure & stable

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Definition:

Linux is an open-source operating system based on UNIX, used to manage system resources and run applications.

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Linux is based on the UNIX operating system. UNIX is a powerful, multi-user, multitasking operating system originally developed in the 1970s at AT&T Bell Labs. It laid the foundation for many modern operating systems, including Linux.

- Linux is free and open-source, accessible to everyone.
- This promotes global collaboration and innovation.
- Linux offers efficient performance and strong security.
- It works well across many devices and industries.

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History of Linux

1: UNIX (1969)

- Developed at Bell Labs
- Multi-user and powerful OS
- But it was not free

2: MINIX (1987)

- Created by Andrew Tanenbaum
- Used for teaching OS concepts
- Limited features

3: Linux (1991)

Created by Linus Torvalds (student from Finland)

- Wanted a free OS like UNIX
- Developed first Linux kernel
- Shared it on internet

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What is a Linux Distribution?

A Linux distribution (distro) is a complete operating system

It includes:

- Linux kernel
- Software
- Tool

Simple meaning:

Linux + programs = Distribution

- Linux alone is not enough
- When we add software, it becomes usable
- That full system is called a distribution

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Examples of Linux Distributions

- Ubuntu → easy for beginners
- Red Hat (RHEL) → used in companies
- CentOS → free server OS
- Debian → very stable
- Fedora → latest features
- Kali Linux → used for security

Around 600 + Linux Distributions are available

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Why many distributions?

Because different users need different things:

- Beginners → easy system
- Companies → stable & secure
- Hackers/security → special tools

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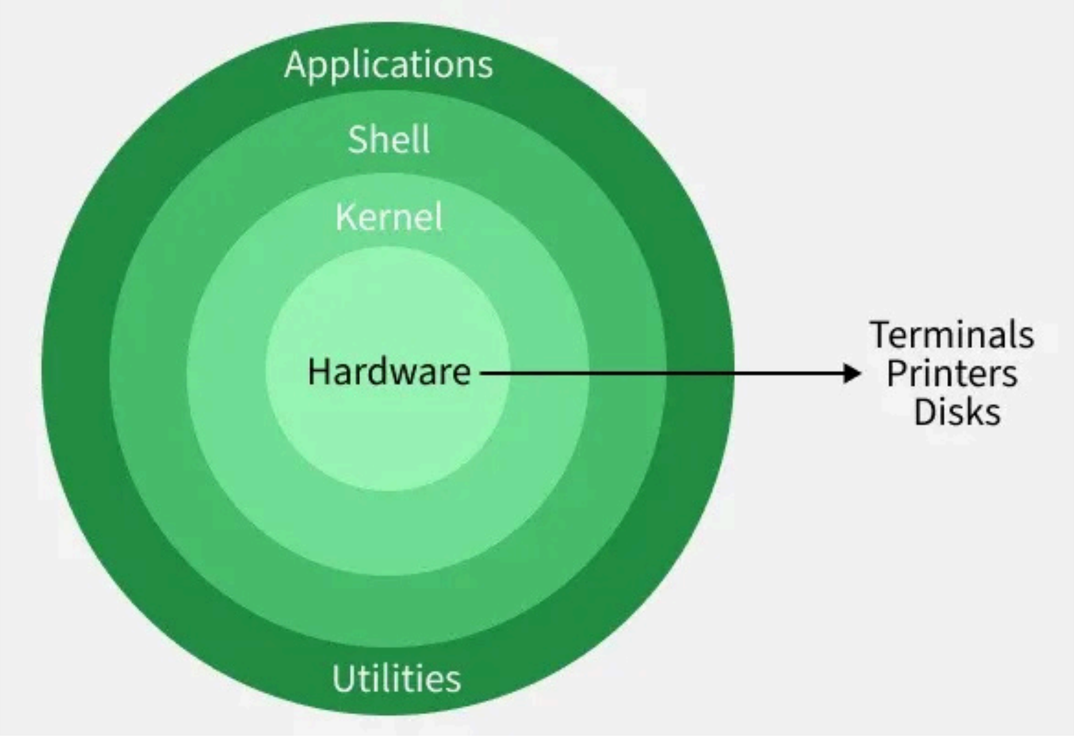
Architecture of Linux:

User → Applications → Shell → System Libraries → Kernel → Hardware



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1.Applications

These are programs used by users

Examples:

- Browser
- Text editor
- Media player

They send requests to the system

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2. Shell

The shell is the command-line interface that allows users to communicate with the operating system by entering commands.

- Interprets and executes user commands.
- Acts as a bridge between user actions and kernel processing.

3. S**ystem Utilities**

System utilities are built-in tools that help users manage, configure, and maintain the operating system.

- Used for tasks like software installation, user management, and monitoring.
- Simplify system administration processes for both beginners and administrators.

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4. Kernel

The kernel is the core of the Linux operating system that manages hardware resources and controls communication between software and hardware.

- Handles process management, memory, and device control.
- Prevents conflicts between multiple running programs.

Types of Kernels: Monolithic, Microkernel, Hybrid, Exokernel

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5. Hardware Layer

The hardware layer consists of physical components that execute commands and provide system resources.

- Includes CPU, RAM, storage, and input/output devices.
- Communicates with the OS using device drivers and kernel services.

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Applications of Linux Operating System:

1.Servers and Hosting

- Powers most web servers, cloud platforms, and data centers worldwide.
- Offers high stability, security, and uptime for critical services.

2. Development

- Provides powerful tools and environments for coding, testing, and debugging.
- Supports multiple programming languages and frameworks.

3. Desktop and Personal Use

- Offers customizable and secure desktop environments.
- Supports daily computing tasks like browsing, office work, and media usage.

4. Cybersecurity

- Widely used for ethical hacking and penetration testing (e.g., Kali Linux).
- Provides advanced tools for threat detection and security analysis.

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Applications of Linux Operating System:

5. Embedded Systems
   - Runs efficiently on low-resource devices like IoT and routers.
   - Supports real-time processing and hardware-level control.
6. Supercomputers
   - Used in high-performance computing environments worldwide.
   - Handles massive scientific calculations and simulations efficiently.
7. Education
   - Helps students learn programming, networking, and system administration.
   - Free and open-source, making it ideal for academic institutions.

Linux Installation